Task 1 – Variables and Console

Create a JavaScript file and:

1. Create variables for:

- name
- age
- city

2. Print them in the console like this:

My name is Sara, I am 22 years old, and I live in Cairo.

Task 2 – Simple Calculator

Create a program that:

1. Creates two numbers.
2. Calculates:

- Sum
- Subtraction
- Multiplication
- Division

3. Print the results in the console.

Example:

Sum = 15
Subtraction = 5
Multiplication = 50
Division = 2

Task 3 – If Condition

Write a program that:

1. Creates a variable age.
2. Check:

- If age is **18 or more** → print "You are an adult"
- Otherwise → print "You are underage"

Task 4 – Even or Odd

Write a program that:

1. Creates a variable number.
2. Check if the number is:

- **Even**
- **Odd**

Print the result in the console.

Task 5 – Loop from 1 to 10

Using a **for loop**:

1. Print numbers from **1 to 10** in the console.

Example:

```
1
2
3
4
...
10
```

Task 6 – Sum of Numbers

Write a program that:

1. Uses a **loop** to calculate the sum of numbers from **1 to 10**.

2. Print the result.

Expected result:

Sum = 55

Task 7 – Array Basics

Create an array called fruits with 5 fruits.

Example:

["apple", "banana", "orange", "mango", "grapes"]

Tasks:

1. Print the first fruit.
2. Print the last fruit.
3. Print the array length.

Task 8 – Loop Through Array

Using a **for loop**:

1. Create an array of **5 student names**.
2. Print each name in the console.

Example:

Student: Ahmed
Student: Sara
Student: Ali
...

Task 9 – Function

Create a function called greetUser.

The function should:

- take name as parameter

- return:

Hello, Ahmed! Welcome to JavaScript.

Call the function and print the result.

Task 10 – Mini Challenge

Create a program that:

1. Create an array of **5 numbers**.
2. Find:

- The **largest number**
- The **smallest number**

Print them in the console.

Example:

Largest number = 50
Smallest number = 3

Task 11 – Logical Operators

Create three variables:

- age
- hasID
- isStudent

Write a program that checks:

- If the user is **18 or older AND has an ID** → print "Access granted"
- Otherwise → print "Access denied"

Use **logical operators (&&, ||, !)**.

Task 12 – Ternary Operator

Create a variable score.

Using a **ternary operator**, check:

- If score is **50 or more** → "Pass"
- Otherwise → "Fail"

Print the result.

Example:

let score = 70;

Output:

Pass

Task 13 – Data Type Coercion

Write a function that prints the result of the following operations and explain the result:

"5" + 2
"5" - 2
true + 1
false + 10

Print the results in the console.

Task 14 – Dialogue Boxes

Create a program that:

1. Ask the user for their **name** using prompt()
2. Show a message using alert():

Example:

Hello Ahmed, welcome to JavaScript!

Task 15 – Loop + Condition

Write a program that:

1. Loop from **1 to 20**
2. If the number is **even**, print:

4 is even

Otherwise print:

5 is odd

Task 16 – User Defined Function

Create a function called calculateArea.

The function should:

- take width and height
- return the **area**

Example:

calculateArea(5, 10)

Output:

Area = 50

Task 17 – Default Parameters

Create a function called greetUser.

- It takes name
- If no name is provided → use "Guest" as a default value

Example:

greetUser()

Output:

Hello Guest

Task 18 – String Methods

Create a variable:

let message = " hello javascript world ";

Using **string methods**, do the following:

1. Remove spaces from the beginning and end
2. Convert the string to **uppercase**
3. Check if the string contains "javascript"

Print the results.

Task 19 – Array Methods

Create an array:

let numbers = [10, 20, 30, 40];

Do the following:

1. Add 50 to the end of the array
2. Remove the first element
3. Print the array
4. Print the array length

Use **array methods**.

Task 20 – Date Object

Write a program that:

1. Create a Date object.
2. Print:
    - Current **year**
    - Current **month**
    - Current **day**

Example output:

Year: 2026
Month: 3
Day: 12